

Vijayabalan J

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PROFESSIONAL SUMMARY

Computer Science and Engineering graduate with practical experience across cloud infrastructure, cybersecurity, system administration, and full-stack development. Built and deployed a production-style auto-scaling AWS infrastructure using IaC. Strong foundation in Linux administration, network security, and software development. Able to design, implement, and document technical systems independently and collaborate across teams.

TECHNICAL SKILLS

Languages: Python, Java, Bash, MySQL

Security: SIEM, Wireshark, Nmap, File Integrity Monitoring

Operating Systems: Linux (Arch, Kali, TrueNAS, Ubuntu), Windows 10/11, macOS

Development: RESTful APIs, MySQL, Spring Data JPA, Postman, AWS (EC2, VPC, ALB, ASG, IAM), Docker, Git, GitHub

PROJECTS

Auto-Scaling Web Infrastructure on AWS

Concepts: Infrastructure as Code, Cloud Architecture, Auto Scaling, High Availability, DevOps, Python Automation

- Provisioned 15 AWS resources using Terraform, including VPC, multi-AZ subnets, IGW, ALB, ASG, Launch Template, Security Groups, and scaling policy.
- Built multi-AZ high availability across ap-south-1a/b using ALB and ASG for traffic distribution and automatic instance replacement.
- Automated EC2 provisioning with cloud-init/UserData, installing Python 3, Flask, virtualenv, and systemd; instances became traffic-ready in ~3 minutes.
- Developed a Python/Boto3 monitoring tool using EC2, ASG, and ELBv2 APIs for real-time infrastructure health monitoring.
- Implemented IAM best practices with a dedicated Terraform IAM user and AWS CLI profile, isolating root access.

Security Operations Center Lab — Wazuh SIEM + AI Integration — Wazuh · Ubuntu · Windows · Ollama · Llama 3.1 8B

Concepts: SIEM, XDR, File Integrity Monitoring, endpoint security, AI-assisted SOC, threat detection

- Deployed Wazuh SIEM/XDR on Ubuntu (VMware) integrated with Windows 10 agent — simulated enterprise SOC environment with real-time endpoint monitoring, threat detection, and incident analysis.
- Implemented File Integrity Monitoring (FIM) for critical system directories, detecting and alerting on unauthorised file creation, modification, and deletion in real time.
- Integrated local LLM (Llama 3.1 8B via Ollama) to automate first-pass security alert triage — translating raw SIEM alerts into structured incident summaries, accelerating analyst investigation workflows.
- Configured secure bidirectional communication over bridged network between Wazuh Manager and Windows agent, validating alerts for authentication anomalies, integrity violations, and endpoint activity.

Self-Hosted Home Server — Private Cloud & NAS — TrueNAS · Nextcloud · Linux · RAID

Concepts: Linux administration, NAS, RAID storage, LAN networking, self-hosted cloud, access control

- Provisioned a TrueNAS-based home server with RAID storage pools and datasets providing redundant private cloud storage eliminating dependency on commercial cloud platforms.
- Deployed Nextcloud for private cloud storage with multi-device file synchronisation; managed user permissions, access controls, and storage quotas via web-based administration.
- Gained practical Linux system administration skills: storage pool management, network configuration, service deployment, and user access management.

EDUCATION

Bachelor of Engineering in Computer Science and Engineering - SRM Valliammai Engineering College | Nov 2022 – Apr 2026 | CGPA 7.6

Coursework: Data Structures and Algorithms, System Architecture, Networking and Cloud Fundamentals

CERTIFICATIONS

AWS Cloud Practitioner Essentials — AWS Skills Builder 2026

Java Foundations — Oracle 2025

Networking Basics — CISCO 2025